SGA 4, Exposé VI

Finiteness Conditions. Fibred Topoi and Sites. Applications to Questions of Passage to the Limit

A. Grothendieck and J.-L. Verdier

0. Introduction

In the present exposé we study two kinds of questions which are closely related.

First, we make a systematic study of finiteness conditions for topoi. We successively study the notions of *quasi-compactness*, *quasi-separatedness*, and *coherence* (quasi-compactness plus quasi-separatedness), inspired by the analogous notions familiar in scheme theory. We treat first the case of an object X of a topos and that of an arrow $u : X \to Y$ in a topos (§1), then the topos E itself (§2), and finally a morphism of topoi $f : E \to E'$ (§3). Section 4 studies these notions for a topos obtained by gluing two topoi (Exposé IV, §9). For the following sections, the essential point to retain from these rather technical developments is the definition of a *coherent topos*: a topos equivalent to a topos of the form $C^{\sim}$, where C is a small site in which finite projective limits are representable and every covering family admits a finite covering subfamily; and the definition of a *coherent morphism* $f : E \to E'$ of coherent topoi: a morphism which, up to equivalence, is associated with a morphism of sites $f : C \to C'$, with C, C' as above and with the corresponding functor $f^* : C' \to C$ left exact.

It is useful to note that, with the finiteness notions used here, and especially with coherent topoi, we deliberately depart, for the first time, from the spaces familiar to analysts and "topologists" in the ordinary sense. For example, the topos associated to a separated topological space X is coherent only if X is a finite set. Thus this exposé is entirely oriented toward the kinds of topoi which arise in algebraic geometry and algebra. In fact, all topoi used so far in these disciplines (except, of course, in algebraic geometry by transcendental methods over the field of complex numbers) are locally coherent.

Second, we develop two *inductive-limit passage theorems* for cohomology of topoi. One says that the functors $H^i(E, -)$ on a coherent topos commute with inductive limits of abelian sheaves. The other says, essentially, that if a topos X is represented as a filtered projective limit of coherent topoi X_i, with coherent transition morphisms $f_{ij} : X_j \to X_i$, then the cohomology of X, with coefficients in an arbitrary abelian sheaf, is computed as the inductive limit of the cohomologies of the X_i. To give this statement a precise meaning, the main point is to define the notion of a *projective limit of topoi*; this is done in §§7–8. It is a useful geometric notion, probably beyond the context of projective limits of schemes, which had served M. Artin as the model for the initial definition. For example, we show how the various notions of "localization at a point" (of a noetherian space, of a scheme for its usual topology, and even for its étale topology) can be unified by defining the localization as the projective limit of the neighborhoods of that point, in perfect accord with the intuitive idea of localization at a point.

The two limit-passage theorems will be restated in the following exposé in the particular framework of étale cohomology of schemes. In that case they are constantly used throughout the rest of the Seminar, and practically everywhere one has worked with étale cohomology. A reader willing to admit these two theorems in the special form in which they are stated in Exposé VII, and interested only in applications to étale cohomology, may therefore omit the present exposé. It is true that the notion of a *constructible object* of a topos, namely an object whose structural morphism $X \to e$ to the final object is coherent, will also play an important role throughout the rest of the Seminar. In the present exposé, however, it plays only a secondary role, if only because in the case of a coherent topos it coincides with the notion of a coherent object, which is the main notion here. The notion is developed independently in Exposé IX for étale sheaves on schemes, and in that setting it has useful special features which underlie the presentation in IX. Since the redaction of IX preceded the present redaction of VI by five years and is nevertheless perfectly usable, we content ourselves at the end of IX with proving the equivalence of the constructibility notion used there with the one introduced here.

1. Finiteness conditions for objects and arrows of a topos

Definition 1.1. Let C be a site. An object X of C is called *quasi-compact* if, for every covering family $(X_i \to X)_{i\in I}$, there exists a finite subset $J \subset I$ such that the subfamily $(X_i \to X)_{i\in J}$ is still covering.

Since a $\mathcal{U}$-topos E is always regarded as a site, this definition applies in particular to objects of E. The following result reduces the site case to the topos case.

Proposition 1.2. Let C be a $\mathcal{U}$-site, let $\epsilon : C \to C^{\sim}$ be the canonical functor, and let X be an object of C. Then X is quasi-compact as an object of the site C if and only if $\epsilon(X)$ is quasi-compact as an object of the topos $C^{\sim}$.

Proof. Assume X quasi-compact. Let $(F_i \to \epsilon(X))_{i\in I}$ be an epimorphic family. For each i, choose an epimorphic family $\epsilon(X_{ij}) \to F_i$. After refining these families, the composites $\epsilon(X_{ij}) \to \epsilon(X)$ may be assumed to come from morphisms $X_{ij} \to X$. Since the family of all $\epsilon(X_{ij}) \to \epsilon(X)$ is epimorphic, the corresponding family $X_{ij} \to X$ is covering. By quasi-compactness of X, a finite subfamily is still covering; hence the corresponding finite subfamily of the $F_i \to \epsilon(X)$ is epimorphic. Conversely, if $\epsilon(X)$ is quasi-compact, any covering of X gives an epimorphic covering of $\epsilon(X)$, and the result is immediate.

Thus quasi-compactness in sites reduces to quasi-compactness in topoi.

Proposition 1.3. Let C be a site and let $(X_i \to X)_{i\in I}$ be a finite covering family whose X_i are quasi-compact. Then X is quasi-compact.

Proof. Using 1.2, reduce to the case where C is a topos. If $(X'_j \to X)_{j\in J}$ is an epimorphic family, then after base change to each X_i it admits a finite epimorphic subfamily. Since I is finite, the union of these finite index sets is finite, and the corresponding subfamily already covers X.

Corollary 1.4. Let $(X_i)_{i\in I}$ be a family of objects of a topos E, and let $X = \coprod_i X_i$. Then X is quasi-compact if and only if every X_i is quasi-compact and all but finitely many X_i are initial.

Proof. The sufficiency follows from 1.3. Conversely, a finite subfamily must cover X, which forces all omitted summands to be initial. To see that each X_i is quasi-compact, write $X \simeq X_i \amalg X_i'$ and extend coverings of X_i to coverings of X.

Remark 1.5.1. Whether an object X of a topos E is quasi-compact depends only on the induced topos $E_{/X}$: it means that the final object of $E_{/X}$ is quasi-compact. Similarly, an object X of a site C is quasi-compact if and only if the final object of the induced site $C_{/X}$ is quasi-compact.

An object X of a topos E is quasi-compact if and only if every filtered increasing family $(X_i)_{i\in I}$ of subobjects of X whose union is X contains an $X_i = X$. This condition depends only on the ordered set of subobjects of X, that is, on the opens of $E_{/X}$.

Let $(X_i)_{i\in I}$ be a generating family of E. From the definitions and 1.3 it follows that, for X to be quasi-compact, it is necessary that X admit a covering by finitely many of the X_i, equivalently that X be a quotient of a finite sum of objects from the generating family. This condition is sufficient if all X_i are quasi-compact. Replacing the family by a subfamily, one may suppose $I \in \mathcal{U}$. Since the set of quotients of an object of E is small, the full subcategory $E_{\mathrm{qc}} \subset E$ of quasi-compact objects is equivalent to a category belonging to $\mathcal{U}$.

Example 1.6.1. Let X be a topological space. An open U is quasi-compact as an object of the site $\mathrm{Open}(X)$ if and only if U is quasi-compact for the induced topology. More generally, a sheaf F on X is quasi-compact as an object of $\mathrm{Top}(X)$ if and only if the étale space associated with F is quasi-compact.

For the category **Sch** of schemes and any of the standard topologies considered later, such as the Zariski, étale, fppf, or fpqc topology, a scheme is quasi-compact for that topology if and only if it is quasi-compact in the usual sense.

Definition 1.7. Let E be a topos and let $f : X \to Y$ be an arrow of E. The arrow f is called *quasi-compact* if, for every arrow $Y' \to Y$ with Y' quasi-compact, the object

$$X' = X \times_Y Y'$$

is quasi-compact. It is called *quasi-separated* if its diagonal

$$X \longrightarrow X \times_Y X$$

is quasi-compact. It is called *coherent* if it is both quasi-compact and quasi-separated.

Spelling out the definition, $f : X \to Y$ is quasi-separated precisely when, for every pair of arrows $g_1, g_2 : X' \rightrightarrows X$ over Y, with X' quasi-compact, the equalizer $\mathrm{Ker}(g_1, g_2)$ is quasi-compact.

Proposition 1.8. Let E be a topos.

(i) Every isomorphism in E is coherent. Composites of quasi-compact, respectively quasi-separated, respectively coherent, morphisms have the same property.

(ii) These three properties are stable under base change.

(iii) If $f : X \to Y$ and $g : Y \to Z$ are morphisms and gf is quasi-separated, then f is quasi-separated.

(iv) If $g : Y \to Z$ is quasi-separated and $gf : X \to Z$ is quasi-compact, respectively coherent, then f is quasi-compact, respectively coherent.

Proof. The proofs are the standard ones from algebraic geometry. For quasi-separatedness of a composite one uses the cartesian square involving the diagonals

$$\begin{tikzcd} X \arrow[d] \arrow[dd, bend right, "\Delta_{X/Z}"'] & \\ X \times_Y X \arrow[r] \arrow[d] & Y \arrow[d, "\Delta_{Y/Z}"] \\ X \times_Z X \arrow[r] & Y \times_Z Y. \end{tikzcd}$$

For (iv), use the graph $\Gamma_f : X \to X \times_Z Y$ and the cartesian diagram comparing it with $\Delta_{Y/Z}$. The remaining part reduces to the fact that a monomorphism is quasi-separated.

Corollary 1.8.1. Every monomorphism is quasi-separated.

Corollary 1.8.2. Let $f : X \to Y$ and $f' : X' \to Y'$ be two S-morphisms in E. If f and f' are quasi-compact, respectively quasi-separated, respectively coherent, then so is

$$f \times_S f' : X \times_S X' \longrightarrow Y \times_S Y'.$$

Remark 1.9.1. Let E be any category with fiber products, and let Q be a class of objects stable under isomorphism, playing the role of quasi-compact objects. One may define Q-quasi-compact, Q-quasi-separated, and Q-coherent morphisms by the preceding formulas. Proposition 1.8 and Corollary 1.8.1 remain valid; so does Corollary 1.8.2 when binary products exist.

Let S be an object of E, and let $f : X \to Y$ be a morphism in $E_{/S}$. By 1.5, f is quasi-compact, quasi-separated, or coherent in $E_{/S}$ if and only if the underlying morphism in E has the same property. In particular, an arrow $f : X \to Y$ is quasi-compact, quasi-separated, or coherent precisely when the structural morphism of the object (X, f) of $E_{/Y}$ to the final object is so. One also says that X is quasi-compact, quasi-separated, or coherent *over* Y.

An object X of a topos E is called *constructible* if it is coherent over the final object. Every morphism from a constructible object to another is coherent. The full subcategory E_{cons} of constructible objects is stable under finite projective limits.

The notions in 1.7 are useful mainly for topoi admitting a generating family of quasi-compact objects. Under this assumption they are local on the base.

Proposition 1.10. Assume E has a generating subcategory C consisting of quasi-compact objects, and let $f : X \to Y$ be a morphism in E.

(i) f is quasi-compact if and only if, for every $S \to Y$ with $S \in \mathrm{Ob}\, C$, the object $X \times_Y S$ is quasi-compact.

(ii) If $(Y_i \to Y)_{i \in I}$ is a covering family, then f is quasi-compact, respectively quasi-separated, respectively coherent, if and only if all base changes

$$f_i : X_i = X \times_Y Y_i \longrightarrow Y_i$$

have the corresponding property.

Proof. For (i), reduce testing against an arbitrary quasi-compact $Y' \to Y$ to a finite cover of Y' by objects of C, and use 1.3. For (ii), it suffices to treat quasi-compactness, since quasi-separatedness is the quasi-compactness of the diagonal and coherence is the conjunction. Again reduce to testing after a finite cover by objects of C.

Corollary 1.11. Let $g : Y \to Z$ be a morphism in E, and let $(f_i : X_i \to Y)_{i \in I}$ be a covering family with target Y.

(i) If I is finite and all gf_i are quasi-compact, then g is quasi-compact.

(ii) If the f_i are quasi-compact and all gf_i are quasi-separated, then g is quasi-separated. If moreover I is finite and the gf_i are coherent, then g is coherent.

Corollary 1.12. Let $(X_i)_{i \in I}$ be a finite family of objects of E, let $X = \coprod_i X_i$, and let $f_i : X_i \to Y$ be morphisms with associated $f : X \to Y$. Then f is quasi-compact, respectively quasi-separated, respectively coherent, if and only if the f_i have the corresponding property. In the quasi-separated case, the finiteness hypothesis on I may be omitted.

Definition 1.13. An object X of a topos E is called *quasi-separated* if, for every quasi-compact object S of E, every morphism $S \to X$ is quasi-compact. Equivalently, for quasi-compact S, T and morphisms $S \to X$, $T \to X$, the product $S \times_X T$ is quasi-compact. An object is *coherent* if it is quasi-compact and quasi-separated.

Proposition 1.14. Let $f : X \to Y$ be a morphism in a topos E.

(i) If Y is quasi-compact, then f quasi-compact implies X quasi-compact. If Y is quasi-separated, then X quasi-compact implies f quasi-compact. If Y is coherent, then f is quasi-compact if and only if X is quasi-compact.

(ii) If Y is quasi-compact, respectively quasi-separated, respectively coherent, and f has the same corresponding property, then X has that property.

Proof. The quasi-compact statements are formal from the definitions. For the quasi-separated case in (ii), let S be quasi-compact and $g : S \to X$. Since Y is quasi-separated, fg is quasi-compact; since f is quasi-separated, it follows that g is quasi-compact.

Corollary 1.15. Every subobject of a quasi-separated object is quasi-separated. For a family $(X_i)_{i \in I}$, the sum $X = \coprod_i X_i$ is quasi-separated, respectively coherent, if and only if the X_i are so; in the coherent case, all but finitely many X_i must be initial.

Corollary 1.16. Under the hypotheses of Proposition 1.10(ii), if the Y_i are coherent, then f is quasi-compact if and only if the objects

$$X_i = X \times_Y Y_i$$

are quasi-compact.

Corollary 1.17. Let E admit a generating family of quasi-compact objects, and let $(g_i : Y_i \to Y)_{i \in I}$ be a covering family, respectively finite covering family, with the Y_i coherent. Then Y is quasi-separated, respectively coherent, if and only if every g_i is quasi-compact, equivalently if and only if for all i, j the object

$$Y_i \times_Y Y_j$$

is quasi-compact.

Corollary 1.17.1. Let E be as in 1.17, let X be coherent, let $\mathcal{R} \rightrightarrows X$ be an equivalence relation, and put $Y = X/\mathcal{R}$. The following are equivalent:

$$Y \text{ is coherent}, \qquad X \to Y \text{ is quasi-compact}, \qquad \mathcal{R} \text{ is quasi-compact}.$$

Corollary 1.18. Let E have a generating subcategory C of quasi-compact objects, and let $Y \in E$. Then Y is quasi-separated if and only if every morphism $X \to Y$, with $X \in \operatorname{ob} C$, is quasi-compact.

Corollary 1.19. Let E admit a generating subcategory of quasi-compact objects. Let $(g_i : Y_i \to Y)_{i \in I}$ be a covering family with the Y_i quasi-separated and the g_i quasi-compact. Then Y is quasi-separated.

Proof. For any quasi-compact X and morphism $X \to Y$, after base change to Y_i the objects $X \times_Y Y_i$ are quasi-compact; since Y_i is quasi-separated, the induced morphisms to Y_i are quasi-compact. Proposition 1.10 then gives the result.

Remark 1.20.1. A reader accustomed to the terms quasi-compact and quasi-separated in algebraic geometry might expect further relations between the absolute notions for objects of E and the relative notions for arrows, such as: if $f : X \to Y$ has quasi-compact separated source X, then f is quasi-separated. Such properties require restrictive finiteness hypotheses on E, stronger than those in Proposition 1.10; they will be studied in the next section.

It again follows from 1.5 that quasi-separatedness of an object X depends only on the induced topos $E_{/X}$; it means that the final object of $E_{/X}$ is quasi-separated, or equivalently that in $E_{/X}$ the product of two quasi-compact objects is quasi-compact.

1.21. The case of sites

Let C be a $\mathcal{U}$-site, E the topos of $\mathcal{U}$-sheaves on C, f an arrow of C, and X an object of C. We say that f is a *quasi-compact morphism* (respectively quasi-separated morphism) of the site C if $\epsilon(f)$ is a quasi-compact (respectively quasi-separated) morphism of the topos E. Similarly, X is called a *quasi-separated object of the site* C if $\epsilon(X)$ is quasi-separated in E. When C is a $\mathcal{U}$-topos with its canonical topology, $\epsilon : C \to E$ is an equivalence, so this agrees with the preceding definitions.

The definition is unchanged when replacing $\mathcal{U}$ by a larger universe $\mathcal{V}$, at least when C has a topologically generating family of quasi-compact objects. Indeed, one reduces to $\mathcal{U} \subset \mathcal{V}$, so that E is a $\mathcal{V}$-site, and compares the corresponding $\mathcal{V}$-topos E' of sheaves on E. The quasi-compact-object case is already known from 1.2; quasi-compactness of arrows follows from the criterion 1.10(i) applied in E and in E' to the same quasi-compact generating family, quasi-separatedness of arrows follows by passing to diagonals, and quasi-separatedness of objects follows from the criterion 1.18.